Motivation Letter (Draft)

To: PhD Admissions Committee — Brain, Mind and Computer Science (BMCS) **Coordinator:** Prof. Anna Spagnolli *(⚠️ verify current coordinator on the official call PDF)* **University of Padova** **Applicant:** Nauval Zulfikar — zulfikar.nauval1998@gmail.com · nauvalzulfikar.vercel.app

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Dear Members of the Committee,

I am applying to the PhD in Brain, Mind and Computer Science because the question I most want to spend three years on lives exactly where this program sits — between computer science and the study of human judgement. I work on language models, and the problem that keeps pulling at me is not whether a model is accurate but *what it has internally come to represent, and how that relates to how a person would judge the same text.*

That interest is grounded in real work, not aspiration. For my MSc dissertation at Aston University (First Class), I fine-tuned the DeBERTa-v3 transformer for multi-aspect classification of multi-tier supplier reviews and, crucially, evaluated the model's **emergent biases against literature baselines** — I was already asking what regularities the model had absorbed, not just whether it scored well. I have since built a retrieval-augmented system over a curated behavioural-literature knowledge base, designed deliberately for reproducibility and ablation, and I treat behaviour itself as something to be modelled: a 450-run agent-based parameter sweep and discrete-choice (MNL, Mixed-Logit) modelling are part of my toolkit. Earlier, leading NLP and model-validation work in a licensed bank gave me a lasting concern with **post-deployment model behaviour** — what models actually do once they are out in the world.

BMCS attracts me precisely because it does not force me to choose between these threads. My proposed project asks how a fine-tuned transformer internally represents *aspect* and *sentiment*, and where those internal representations align with or diverge from human annotation — a question that is at once an interpretability problem (probing and intervening on model internals) and a cognitive one (comparing machine and human evaluative judgement). The program's interdisciplinary framing, and its attention to human-AI interaction, is the right home for it.

I want to be candid about fit. My research background is in applied machine learning and behavioural modelling, not cognitive neuroscience or experimental psychology — this is a genuine interdisciplinary stretch for me. I am not claiming expertise I do not have. What I bring is demonstrable strength on the computational side — transformer fine-tuning, representation and bias evaluation, reproducible experimentation — and a real eagerness to learn the cognitive-science methods properly from the program's faculty. I have scoped my project so that the part I contribute is the part I can credibly deliver, with the cognitive theory supplied by Padova.

I am an Indonesian national, currently relocatable within Europe, eligible for the EU Researcher (Hosting Agreement) route, and my MSc was taught entirely in English. I would be grateful for the opportunity to discuss how my interpretability-meets-judgement angle fits the program's direction.

Thank you for your consideration.

Sincerely, **Nauval Zulfikar** zulfikar.nauval1998@gmail.com · +44 7300 469048

> ⚠️ *Draft ~390 words. Grounded only in the verified profile. Coordinator name and any faculty/paper references to be verified before sending. Honest about the secondary/interdisciplinary-stretch fit per the task brief.*